

Lecture 5 — Linear Algebra

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Morning challenge!

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- ③ For $x = (1, 2, 3, 4, 5)$ and $y = (1.5, 4, 4, 9, 14)$

- ① Calculate β and α and for the OLS regression line using the formulas:

$$\hat{\beta} = \frac{\sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})}{\sum_{i=1}^n (X_i - \bar{X})^2}$$

$$\hat{\alpha} = \bar{Y} - \hat{\beta}\bar{X}$$

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- Substitution: solve for one variable — in terms of the others — and plug the result into other equations
- Elimination: manipulate equations by applying operations to both the left and right hand side, and combine equations to reduce them

Substitution Example

$$x + y = 5 \quad (1)$$

$$3x - 2y = 5 \quad (2)$$

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$$3(5 - y) - 2y = 5$$

$$15 - 5y = 5$$

$$-5y = -10$$

$$y = 2$$

- Substitute y for 2 in equation (1)

$$x + 2 = 5$$

$$x = 3$$

Elimination Example

$$2x - y + 3z = 9 \quad (3)$$

$$x + 4y - 5z = -6 \quad (4)$$

$$x - y + z = 2 \quad (5)$$

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$$-2(x + 4y - 5z) = -2(-6)$$

$$-2x - 8y + 10z = 12$$

- Transform equation (5) so that, when summed with equation (3), x is eliminated

$$-2(x - y + z) = -2(2)$$

$$-2x + 2y - 2z = -4$$

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$$\begin{aligned}2x - y + 3z &= 9 \\ -2x + 2y - 2z &= -4 \\ y + z &= 5\end{aligned}$$

Elimination Example

- Use our new equations to eliminate y

$$-9y + 13z = 21$$

$$9(y + z) = 9(5)$$

$$22z = 66$$

$$z = 3$$

Elimination Example

- Use our new equations to eliminate y

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- Use $z = 3$ to solve for y through substitution

$$-9y + 13(3) = 21$$

$$-9y = -18$$

$$y = 2$$

Elimination Example

- Use $y = 2$ and $z = 3$ to solve for x

$$2x - (2) + 3(3) = 9$$

$$2x = 2$$

$$x = 1$$

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 - A dataset is an example of many more equations (observations) than unknowns (variables)
- Linear algebra helps us solve these complications

Vector and scalars

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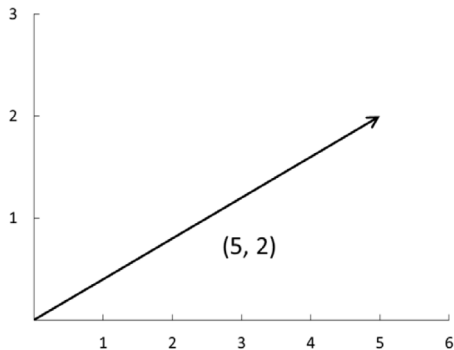


Figure: Vector $(5,2)$

Vector and scalars

- A *scalar* is a single number/element

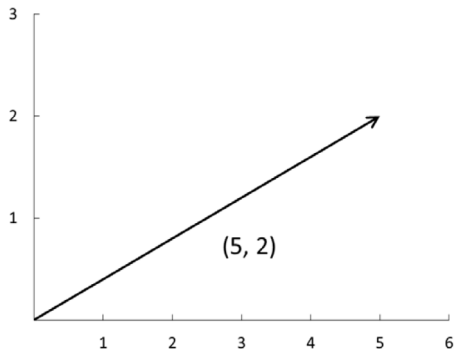


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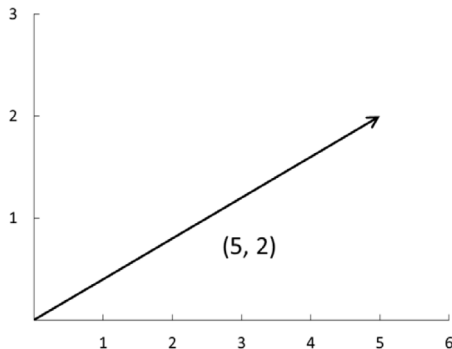


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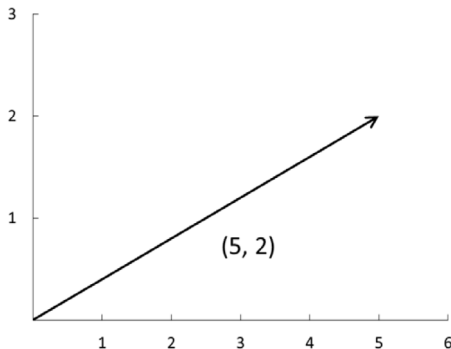


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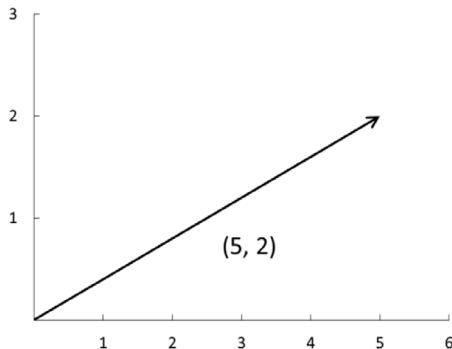


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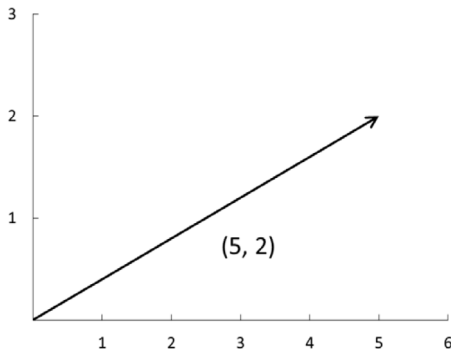


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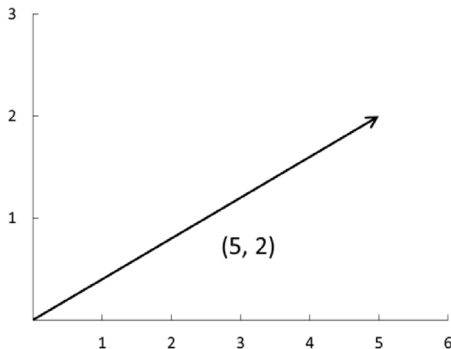


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- For Vector (5,2) length = $\sqrt{5^2 + 2^2} \approx 5.39$
- This can be expanded to
$$\|a\| = \sqrt{a_1^2 + a_2^2 + \dots + a_n^2}$$

Vector addition / subtraction

- **Vector sums**

$$\begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix} + \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix} = \begin{bmatrix} a_1 + b_1 \\ a_2 + b_2 \\ a_3 + b_3 \end{bmatrix}$$

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- **Vector differences**

$$\begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix} - \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix} = \begin{bmatrix} a_1 - b_1 \\ a_2 - b_2 \\ a_3 - b_3 \end{bmatrix}$$

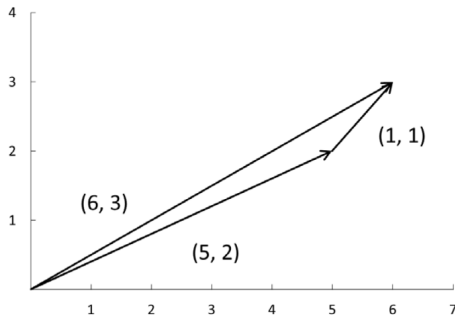


Figure: Vector addition: $(5,2) + (1,1)$

Scalar multiplication and dot product

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- **Scalar multiplication**

$$c \cdot \vec{a} = c \cdot \begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix} = \begin{bmatrix} c \cdot a_1 \\ c \cdot a_2 \\ c \cdot a_3 \end{bmatrix}$$

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- **Dot product (or scalar product)**

$$\begin{bmatrix} a_1 \\ a_2 \end{bmatrix} \cdot \begin{bmatrix} b_1 \\ b_2 \end{bmatrix} = a_1 b_1 + a_2 b_2$$

This requires that the two vectors be of equal dimension.

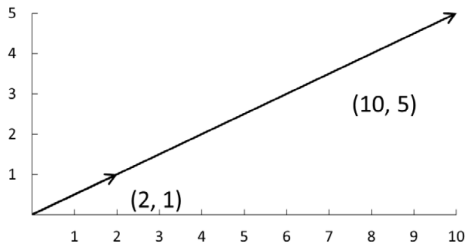


Figure: scalar multiplication: $5a$ where $a=(2,1)$

Dot Product Example

$$\begin{bmatrix} 6 \\ 5 \\ 4 \end{bmatrix} \cdot \begin{bmatrix} 9 \\ 8 \\ 7 \end{bmatrix}$$

$$54 + 40 + 28 = 122$$

Matrices

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Example:

$$A_{3 \times 3} = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$

Important types of matrices

Zero matrix:

$$A_{3 \times 3} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

Diagonal matrix:

$$A_{3 \times 3} = \begin{bmatrix} a_{11} & 0 & 0 \\ 0 & a_{22} & 0 \\ 0 & 0 & a_{33} \end{bmatrix}$$

Identity matrix:

$$I_{1 \times 1} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Symmetric matrix:

$$A_{3 \times 3} = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{12} & a_{22} & a_{23} \\ a_{13} & a_{23} & a_{33} \end{bmatrix}$$

Lower triangular matrix:

$$A_{3 \times 3} = \begin{bmatrix} a_{11} & 0 & 0 \\ a_{21} & a_{22} & 0 \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$

Upper triangular matrix:

$$I_{1 \times 1} = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ 0 & a_{22} & a_{23} \\ 0 & 0 & a_{33} \end{bmatrix}$$

Matrix transposition

The *transpose* switches the rows and columns of the matrix.
The first row becomes the first column, and so on

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Example:

$$A_{2 \times 3} = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$$

$$A_{3 \times 2}^T = \begin{bmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{bmatrix}$$

Matrix addition, subtraction and scalar multiplication

Matrix addition and subtraction: simply add/subtract each corresponding element!

$$A_{3 \times 3} \pm B_{3 \times 3} = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \pm \begin{bmatrix} b_{11} & b_{12} & b_{13} \\ b_{21} & b_{22} & b_{23} \\ b_{31} & b_{32} & b_{33} \end{bmatrix}$$

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Scalar multiplication:

$$5 \times A = \begin{bmatrix} 5 \times a_{11} & 5 \times a_{12} & 5 \times a_{13} \\ 5 \times a_{21} & 5 \times a_{22} & 5 \times a_{23} \\ 5 \times a_{31} & 5 \times a_{32} & 5 \times a_{33} \end{bmatrix}$$

Matrix Addition Example

$$\begin{bmatrix} 6 & 7 & 8 \\ 5 & 6 & 0 \\ 5 & 0 & 4 \end{bmatrix} + \begin{bmatrix} 0 & 3 & 3 \\ 1 & 4 & 1 \\ 0 & 8 & 7 \end{bmatrix}$$

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$$\begin{bmatrix} 6 & 10 & 11 \\ 6 & 10 & 1 \\ 5 & 8 & 11 \end{bmatrix}$$

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 $A_{n \times m} \cdot B_{m \times p}$
- This will result in a matrix of dimensions $n \times p$
- Therefore, $A \times B$ will not result in the same matrix as $B \times A$
 - Left multiplication: multiply by the matrix on the left
 - Right multiplication: multiply by the matrix on the right

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- Repeat this until all values of matrix C are filled in

Matrix multiplication - example

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$$A \times B = \begin{bmatrix} 9 & 11 \\ 16 & 18 \end{bmatrix}$$

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- A matrix with a non-zero determinant is *nonsingular* and can be *inverted* which is important for solving systems of equations

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- Calculation: “difference of the diagonal products”

Determinant

Determinant for a 2×2 matrix:

$$A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$$

Is given by:

$$|A| = a_{11}a_{22} - a_{12}a_{21}$$

Example:

$$B = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$$

$$|B| = (1 \cdot 4) - (2 \cdot 3) = 4 - 6 = -2$$

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- **Minor**: determinant of a *submatrix*
- **Submatrix** of a matrix value is what's left over when you eliminate the row and column of that value

Determinants of larger matrices

- Finding a specific Minor M_{11} of Matrix A
- Take the submatrix of a_{11}

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$

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Determinants of larger matrices

- Finding a specific Minor M_{11} of Matrix A
- Take the submatrix of a_{11}
- The Minor is the the determinant of the 2×2 submatrix

$$A = \begin{bmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{bmatrix}$$

$$M_{11} = a_{22}a_{33} - a_{23}a_{32}$$

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Determinants of larger matrices

- So now we know how to find **submatrices**, which we take the determinants of to get **Minors**, which are made positive or negative as **cofactors**
- **Laplace Expansion**: determinant of a matrix bigger than 2×2 is the sum of products of each element and its *cofactor* for *any one* row or column
- Since we can use any row or column, we need to calculate 3 cofactors for a 3×3 matrix, multiply them by their respective element in the matrix, and sum the results

Full Example

$$A = \begin{bmatrix} 1 & 4 & 3 \\ 1 & 2 & 0 \\ 2 & 3 & 1 \end{bmatrix}$$

Submatrix for Minor M_{11}

$$A = \begin{bmatrix} \cancel{1} & \cancel{4} & \cancel{3} \\ \cancel{1} & 2 & 0 \\ \cancel{2} & 3 & 1 \end{bmatrix}$$

Calculate Determinant

$$A = \begin{bmatrix} 2 & 0 \\ 3 & 1 \end{bmatrix}$$

$$(2 \cdot 1) - (0 \cdot 3) = 2$$

Determine cofactor sign

- The sign of this cofactor C_{11} will be positive ($C_{11} = 2$)

$$\begin{bmatrix} a_{11}(+) & a_{12}(-) & a_{13}(+) \\ a_{21}(-) & a_{22}(+) & a_{23}(-) \\ a_{31}(+) & a_{32}(-) & a_{33}(+) \end{bmatrix}$$

Cofactor C_{12}

$$A = \begin{bmatrix} \cancel{1} & \cancel{4} & \cancel{3} \\ 1 & \cancel{2} & 0 \\ 2 & \cancel{3} & 1 \end{bmatrix}$$

$$M_{12} = (1 \cdot 1) - (0 \cdot 2) = 1$$

- The sign of this cofactor will be negative ($C_{12} = -1$)

Cofactor C_{13}

$$A = \begin{bmatrix} \cancel{1} & \cancel{4} & \cancel{3} \\ 1 & 2 & 0 \\ 2 & 3 & \cancel{1} \end{bmatrix}$$

$$M_{13} = (1 \cdot 3) - (2 \cdot 2) = -1$$

- The sign of this cofactor will be positive ($C_{13} = -1$)

Bringing it all together

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$$|A| = 1(2) + 4(-1) + 3(-1) = -5$$

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This result is replicable for *any* row or column

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Deriving Inverse Matrix for 2×2

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Deriving Inverse Matrix for 2×2

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- Final step is to multiply C^T by $\frac{1}{|A|}$

Example: Inverting 2×2 Matrix

- Matrix $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$

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- What is $A \times A^{-1}$?

Example: Inverting 3×3 Matrix

- Same steps apply
- Challenge will be to create the matrix of cofactors (C)

Start by calculating each Minor

$$A = \begin{bmatrix} 1 & 2 & 1 \\ 0 & 4 & 3 \\ -6 & -2 & 2 \end{bmatrix}$$

Start by calculating each Minor

$$A = \begin{bmatrix} 1 & 2 & 1 \\ 0 & 4 & 3 \\ -6 & -2 & 2 \end{bmatrix}$$

$$M_{11} = \begin{vmatrix} 4 & 3 \\ -2 & 2 \end{vmatrix} =$$
$$(4 \times 2) - (3 \times -2) = 14$$

Start by calculating each Minor

$$A = \begin{bmatrix} 1 & 2 & 1 \\ 0 & 4 & 3 \\ -6 & -2 & 2 \end{bmatrix}$$

$$M_{11} = \begin{vmatrix} 4 & 3 \\ -2 & 2 \end{vmatrix} =$$
$$(4 \times 2) - (3 \times -2) = 14$$

$$M_{21} = \begin{vmatrix} 2 & 1 \\ -2 & 2 \end{vmatrix} =$$
$$(2 \times 2) - (1 \times -2) = 6$$

Start by calculating each Minor

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$$M_{11} = \begin{vmatrix} 4 & 3 \\ -2 & 2 \end{vmatrix} =$$
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$$(2 \times 2) - (1 \times -2) = 6$$

$$M_{31} = \begin{vmatrix} 2 & 1 \\ 4 & 3 \end{vmatrix} =$$
$$(2 \times 3) - (1 \times 4) = 2$$

Start by calculating each Minor

$$A = \begin{bmatrix} 1 & 2 & 1 \\ 0 & 4 & 3 \\ -6 & -2 & 2 \end{bmatrix}$$

$$M_{11} = \begin{vmatrix} 4 & 3 \\ -2 & 2 \end{vmatrix} =$$
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$$M_{31} = \begin{vmatrix} 2 & 1 \\ 4 & 3 \end{vmatrix} =$$
$$(2 \times 3) - (1 \times 4) = 2$$

$$M_{12} = \begin{vmatrix} 0 & 3 \\ -6 & 2 \end{vmatrix} =$$
$$(0 \times 2) - (3 \times -6) = 18$$

Start by calculating each Minor

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$$(1 \times 2) - (1 \times -6) = 8$$

Start by calculating each Minor

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$$(1 \times 3) - (1 \times 0) = 3$$

Start by calculating each Minor

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$$(1 \times 3) - (1 \times 0) = 3$$

$$M_{13} = \begin{vmatrix} 0 & 4 \\ -6 & -2 \end{vmatrix} =$$
$$(0 \times -2) - (4 \times -6) = 24$$

Start by calculating each Minor

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$$M_{23} = \begin{vmatrix} 2 & 1 \\ 4 & 3 \end{vmatrix} =$$
$$(2 \times 3) - (1 \times 4) = 10$$

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$$M_{33} = \begin{vmatrix} 1 & 2 \\ 0 & 4 \end{vmatrix} =$$
$$(1 \times 4) - (2 \times 0) = 4$$

This gives us the Cofactor Matrix C

$$A = \begin{bmatrix} 1 & 2 & 1 \\ 0 & 4 & 3 \\ -6 & -2 & 2 \end{bmatrix}$$

$$C = \begin{bmatrix} 14 & -18 & 24 \\ -6 & 8 & -10 \\ 2 & -3 & 4 \end{bmatrix}$$

This gives us the Cofactor Matrix C

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$$C = \begin{bmatrix} 14 & -18 & 24 \\ -6 & 8 & -10 \\ 2 & -3 & 4 \end{bmatrix}$$

- Determinant $|A|$ is the dot product of *any* row or column of elements and their respective cofactors
- Using row 1, $|A| = (1)(14) + (2)(-18) + (1)(24) = 2$

Take the transpose of the Cofactor Matrix C^T , multiply by inverse determinant $\frac{1}{|A|}$

$$C = \begin{bmatrix} 14 & -18 & 24 \\ -6 & 8 & -10 \\ 2 & -3 & 4 \end{bmatrix}$$

$$C^T = \begin{bmatrix} 14 & -6 & 2 \\ -18 & 8 & -3 \\ 24 & -10 & 4 \end{bmatrix}$$

Take the transpose of the Cofactor Matrix C^T , multiply by inverse determinant $\frac{1}{|A|}$

$$C = \begin{bmatrix} 14 & -18 & 24 \\ -6 & 8 & -10 \\ 2 & -3 & 4 \end{bmatrix}$$

$$C^T = \begin{bmatrix} 14 & -6 & 2 \\ -18 & 8 & -3 \\ 24 & -10 & 4 \end{bmatrix}$$

$$A^{-1} = \frac{1}{|A|} C^T = \frac{1}{2} \begin{bmatrix} 14 & -6 & 2 \\ -18 & 8 & -3 \\ 24 & -10 & 4 \end{bmatrix} = \begin{bmatrix} 7 & -3 & 1 \\ -9 & 4 & -\frac{3}{2} \\ 12 & -5 & 2 \end{bmatrix}$$

Did it work?

$$A \begin{bmatrix} 1 & 2 & 1 \\ 0 & 4 & 3 \\ -6 & -2 & 2 \end{bmatrix} \times A^{-1} \begin{bmatrix} 7 & -3 & 1 \\ -9 & 4 & -\frac{3}{2} \\ 12 & -5 & 2 \end{bmatrix}$$

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$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Matrix and vector properties

Table 12.2: Matrix and Vector Transpose Properties

Inverse	$(A^T)^T = A$
Additive property	$(A + B)^T = A^T + B^T$
Multiplicative property	$(AB)^T = B^T A^T$
Scalar multiplication	$(cA)^T = cA^T$
Inverse transpose	$(A^{-1})^T = (A^T)^{-1}$
If A is symmetric	$A^T = A$

Table 12.3: Matrix Determinant Properties

Transpose property	$\det(A) = \det(A^T)$
Identity matrix	$\det(I) = 1$
Multiplicative property	$\det(AB) = \det(A) \det(B)$
Inverse property	$\det(A^{-1}) = \frac{1}{\det(A)}$
Scalar multiplication ($n \times n$)	$\det(cA) = c^n \det(A)$
If A is triangular or diagonal	$\det(A) = \prod_{i=1}^n a_{ii}$

Table 12.4: Matrix Inverse Properties

Inverse	$(A^{-1})^{-1} = A$
Multiplicative property	$(AB)^{-1} = B^{-1} A^{-1}$
Scalar multiplication ($n \times n$)	$(cA)^{-1} = c^{-1} A^{-1}$ if $c \neq 0$

Linear Independence

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A set of vectors is *linearly independent* if we cannot write any vector in the set as a combination of other vectors in the set.

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Examples:

- Suppose $v_1 = (1, 3)$ and $v_2 = (3, 9)$. These vectors are not linearly independent because $3v_1 - v_2 = 0$
- Suppose $v_1 = (1, 3)$ and $v_2 = (2, 9)$. These vectors are linearly independent because the only a_i that allow $a_1v_1 - a_2v_2 = 0$ are $a_i = 0$

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Linearly independent matrices have non-zero determinants, can be inverted

Linearly dependent matrices have a determinant of zero, cannot be inverted. In statistics, this is called *multicollinearity*

Matrix rank

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 - Linear dependence is called *multicollinearity* in a statistics context

Systems of equations

How determined is your system of equations?

- Uniquely determined
 - Same number of equations and variables to solve for.
 - Yields one unique solution
- Overdetermined
 - More equations than unknowns
 - Equations may be contradictory
- Undetermined
 - More unknown than equations
 - Can occur if equations are not linearly independent - each equation must give us new information if we want to solve the system.
 - Infinite number of possible solutions

Systems of Equations as Matrices

$$2x - y + 3z = 9$$

$$x + 4y - 5z = -6$$

$$x - y + z = 2$$

becomes...

$$\begin{bmatrix} 2 & -1 & 3 \\ 1 & 4 & -5 \\ 1 & -1 & 1 \end{bmatrix} \times \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 9 \\ -6 \\ 2 \end{bmatrix}$$

Solving systems of equations - Matrix inversion

Steps:

Solving systems of equations - Matrix inversion

Steps:

- 1 Arrange the equations in the format $Ax = c$

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- 3 Calculate A^{-1}

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 - Adjoint matrix = transpose of the matrix composed of the determinants of each minor
- ④ Multiply A^{-1} by the vector of constants (c)

Solving systems of equations - Matrix inversion

Steps:

- ① Arrange the equations in the format $Ax = c$
- ② Check the determinant of A . If it is non-zero, can be inverted.
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 - Inverse = $1/\text{determinant}$ times adjoint matrix
 - Adjoint matrix = transpose of the matrix composed of the determinants of each minor
- ④ Multiply A^{-1} by the vector of constants (c)
 - Why? We want to isolate matrix x , so we need to multiply both sides of the equation by the inverse of matrix A (divide both sides by matrix A)

Matrix inversion example

$$\begin{bmatrix} 2 & -1 & 3 \\ 1 & 4 & -5 \\ 1 & -1 & 1 \end{bmatrix} \times \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 9 \\ -6 \\ 2 \end{bmatrix}$$

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- Take the determinant $|A|$ using the Laplace Expansion
 - Pick a row or column
 - Find the determinants of three submatrices
 - Multiply each element of our chosen row/column by those determinants, and sum
- $|A| = -11$
- Non-zero, so we're good to continue!

Matrix inversion example

$$\begin{bmatrix} 2 & -1 & 3 \\ 1 & 4 & -5 \\ 1 & -1 & 1 \end{bmatrix} \times \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 9 \\ -6 \\ 2 \end{bmatrix}$$

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- Calculate the adjoint matrix C^T

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$$C^T = \begin{bmatrix} -1 & -2 & -7 \\ -6 & -1 & 13 \\ -5 & 1 & 9 \end{bmatrix}$$

Matrix inversion example

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- Multiply the inverse determinant $\frac{1}{|A|}$ by the adjoint matrix C^T

Matrix inversion example

$$C^T = \begin{bmatrix} -1 & -2 & -7 \\ -6 & -1 & 13 \\ -5 & 1 & 9 \end{bmatrix}$$

- Calculate the inverse matrix A^{-1}
- Multiply the inverse determinant $\frac{1}{|A|}$ by the adjoint matrix C^T

$$A^{-1} = \frac{1}{-11} \begin{bmatrix} -1 & -2 & -7 \\ -6 & -1 & 13 \\ -5 & 1 & 9 \end{bmatrix}$$

Matrix inversion example

Last step: Multiply A^{-1} by the vector of constants (c)

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \frac{1}{-11} \begin{bmatrix} -1 & -2 & -7 \\ -6 & -1 & 13 \\ -5 & 1 & 9 \end{bmatrix} \times \begin{bmatrix} 9 \\ -6 \\ 2 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$$

Overdetermined systems of equations

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- **Overdetermined**: more linearly independent equations than unknowns, so there is no solution (multiple contradicting solutions)

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$$0 = 0m + b$$

$$2 = 1m + b$$

$$1 = 2m + b$$

$$5 = 3m + b$$

$$3 = 4m + b$$

$$2 = 5m + b$$

$$4 = 6m + b$$

$$5 = 7m + b$$

$$5 = 8m + b$$

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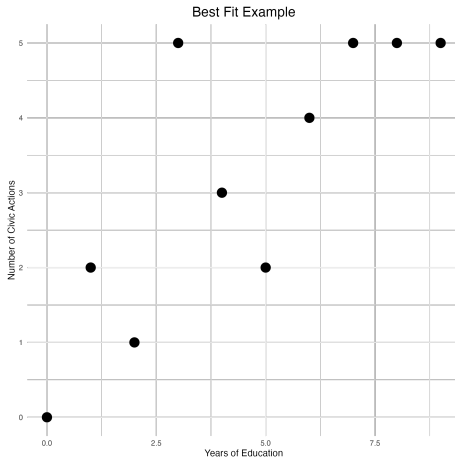
- **Overdetermined:** more linearly independent equations than unknowns, so there is no solution (multiple contradicting solutions)
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$$\bullet \begin{bmatrix} 0 & 1 \\ 1 & 1 \\ 2 & 1 \\ 3 & 1 \\ 4 & 1 \\ 5 & 1 \\ 6 & 1 \\ 7 & 1 \\ 8 & 1 \\ 9 & 1 \end{bmatrix} \times \begin{bmatrix} m \\ b \end{bmatrix} = \begin{bmatrix} 0 \\ 2 \\ 1 \\ 5 \\ 3 \\ 2 \\ 4 \\ 5 \\ 5 \\ 5 \end{bmatrix}$$

- We'll call these matrices X , $\hat{\beta}$, and Y

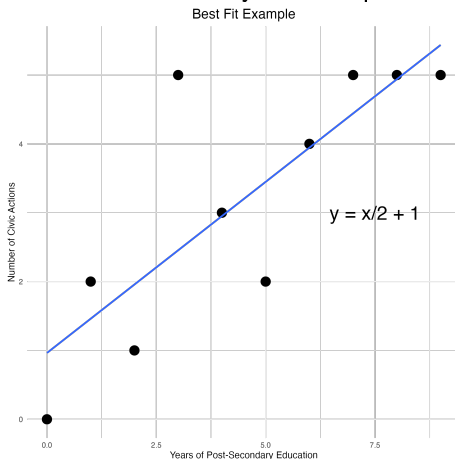
This is an overdetermined problem

The matrix on the last slide represents a series of points



No line goes through all points, calculate the "best fit"

This "best fit line" is an "Ordinary Least Squares regression line"



Solving for Best Fit line with matrices

- We can find the slope m and intercept b of the best fit line using matrix algebra!

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- $X^T X \hat{\beta} = X^T Y$ provides a best fit solution to $X \hat{\beta} = Y$
- $\hat{\beta} = (X^T X)^{-1} X^T Y$

Solving for Best Fit line with matrices

- If we multiply matrix $X^T \times X$, we get $\begin{bmatrix} 285 & 45 \\ 45 & 10 \end{bmatrix}$
- How do we take the inverse of that?
 - Determinant $|X^T X| = (285 * 10) - (45 * 45) = 825$

Solving for Best Fit line with matrices

- If we multiply matrix $X^T \times X$, we get $\begin{bmatrix} 285 & 45 \\ 45 & 10 \end{bmatrix}$
- How do we take the inverse of that?
 - Determinant $|X^T X| = (285 * 10) - (45 * 45) = 825$
 - Adjoint matrix $C^T = \begin{bmatrix} 10 & -45 \\ -45 & 285 \end{bmatrix}$

Solving for Best Fit line with matrices

- If we multiply matrix $X^T \times X$, we get $\begin{bmatrix} 285 & 45 \\ 45 & 10 \end{bmatrix}$
- How do we take the inverse of that?
 - Determinant $|X^T X| = (285 * 10) - (45 * 45) = 825$
 - Adjoint matrix $C^T = \begin{bmatrix} 10 & -45 \\ -45 & 285 \end{bmatrix}$
 - $\frac{1}{825} \begin{bmatrix} 10 & -45 \\ -45 & 285 \end{bmatrix} = \begin{bmatrix} \frac{10}{825} & \frac{-45}{825} \\ \frac{-45}{825} & \frac{285}{825} \end{bmatrix}$
- Multiply by the product of C^T and $Y = \begin{bmatrix} 185 \\ 32 \end{bmatrix}$

Solving for Best Fit line with matrices

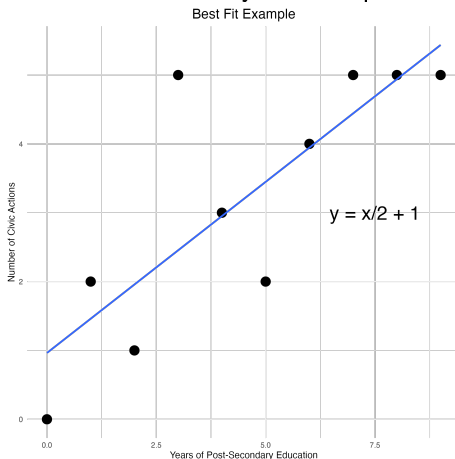
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Solving for Best Fit line with matrices

- $\hat{\beta} = (X^T X)^{-1} X^T Y$
- $\hat{\beta} = \begin{bmatrix} \frac{10}{825} & \frac{-45}{825} \\ \frac{-45}{825} & \frac{285}{825} \end{bmatrix} \times \begin{bmatrix} 185 \\ 32 \end{bmatrix} = \begin{bmatrix} \approx 0.5 \\ \approx 1.0 \end{bmatrix}$

No line goes through all points, calculate the "best fit"

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Challenge!

- Find $C = \begin{bmatrix} 4 & 2 \\ -\frac{2}{3} & -1 \end{bmatrix} + \begin{bmatrix} 3 & 43 \\ -4 & 3 \end{bmatrix}$
- Suppose $A = \begin{bmatrix} 3 & 6 \\ -4 & -8 \end{bmatrix}$ and $B = \begin{bmatrix} -10 & -4 \\ 5 & 2 \end{bmatrix}$. Find AB and BA . Is $AB = BA$?
- Find x , y and z : $\begin{bmatrix} 3 & 1 & 4 \\ 1 & 5 & x \\ 2 & 6 & 5 \end{bmatrix} \begin{bmatrix} 2 & 7 & 1 \\ 8 & 2 & 8 \\ 1 & 8 & y \end{bmatrix} = \begin{bmatrix} z & 55 & 19 \\ 51 & 89 & 59 \\ 57 & 66 & 60 \end{bmatrix}$
- Suppose $A = \begin{bmatrix} 4 & -2 \\ 3 & -1 \end{bmatrix}$. Find A^{-1}